

# Selection on Exit and Sticky Prices

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# Turnover, selection, and monetary policy

- Markets renew constantly: about 8% of products turn over each quarter ([Argente et al., 2018](#)).
  - Turnover and taste shifts move measured annual inflation by 2.6pp ([Braun and Lein, 2021](#)).
- Exit is selective.
  - Literature prices this selection in productivity units ([Caballero and Hammour, 1994](#); [Foster et al., 2016](#)).

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- Exit is selective.
  - Literature prices this selection in productivity units ([Caballero and Hammour, 1994](#); [Foster et al., 2016](#)).
- What does this selection do to the real effects of a nominal shock?

Does the composition of entry and exit matter for the transmission of aggregate shocks?

# This paper

- I price the composition of turnover in terms of real effects of nominal shocks.
- I document stylized facts in scanner data linking product exit, size, and price setting.
- Menu-cost model with firm heterogeneity, aggregate risk, and state-dependent exit and entry.
  - Experiment: hold the total exit rate fixed, make turnover random.
  - Selection raises the one-year cumulative output response by 15%,
  - by almost 35% in low-inflation regimes.
  - Analytical model: a sufficient condition for selection to amplify.
- The result extends to time-dependent pricing: 2.4% under a Calvo lottery.

**Selection on exit amplifies the real effects of aggregate nominal shocks.**

## Related literature

- Size of monetary non-neutrality in menu-cost economies.  
[Golosov and Lucas \(2007\)](#); [Nakamura and Steinsson \(2010\)](#); [Midrigan \(2011\)](#); [Blanco et al. \(2024\)](#)  
↪ The pricing block stays as in Golosov and Lucas; amplification comes from who remains.
- Firm turnover and nominal rigidities.  
[Bilbiie \(2021\)](#); [Oikawa and Ueda \(2015\)](#); [Read \(2020\)](#); [Hamano and Zanetti \(2022\)](#)  
↪ These papers move the number of producers; my exiting firms carry sticky prices out.
- State- vs. time-dependent pricing.  
[Alvarez et al. \(2016, 2017, 2022\)](#); [Auclert et al. \(2024\)](#)  
↪ Selection amplifies in both classes, through different channels.
- Nonlinear monetary transmission.  
[Gagnon \(2009\)](#); [Vavra \(2014\)](#); [Alvarez et al. \(2019\)](#); [Karadi and Reiff \(2019\)](#)  
↪ The amplification from selection is regime dependent.

# Roadmap

- 1 Introduction
- 2 Empirical evidence
- 3 Model
- 4 Low and high inflation regimes
- 5 Conclusion

- One product characteristics predicts both selection and pricing behavior:
  - Product revenues.
- Dutch retail scanner data, 2008–2020 (AiMark): fast-moving consumer goods.
  - 163.5 million non-promotion transactions, ~1M UPCs, 719 categories, 231 chains.
  - About €1.8bn of sector revenue in 2008, out of €30bn.
  - UPC × chain × province; monthly posted price.
- Monthly exit rate 2.0%, price-change frequency 9.0%.
- Product exit: a year-long losing revenue streak, and national (82%, revenue-weighted).

► Filters

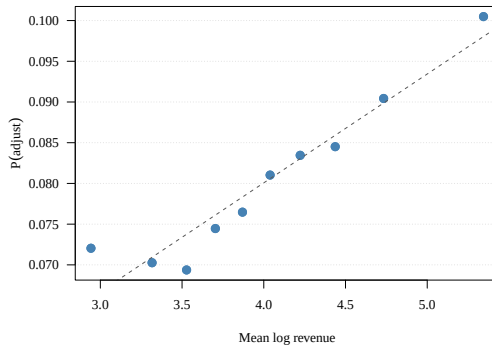
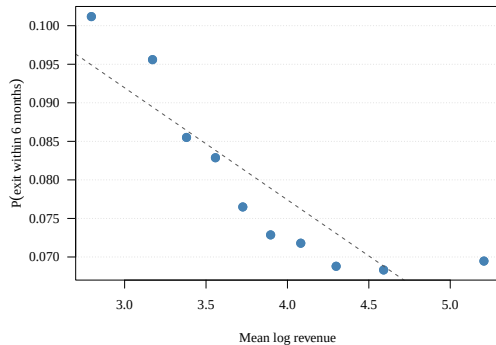
► Descriptives

► Whose prices?

► Pre-exit revenue

► Exit types

## Low-revenue products exit more and adjust less



► Size of changes

► Robustness

# Model

- Household: log utility, complete markets.
- Nominal spending target  $\log S_t = \bar{\pi} + \log S_{t-1} + \sigma_S \eta_t$ .
- Final good: CES over an endogenous measure  $\mu_t$  of varieties.

# Model

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- Nominal spending target  $\log S_t = \bar{\pi} + \log S_{t-1} + \sigma_S \eta_t$ .
- Final good: CES over an endogenous measure  $\mu_t$  of varieties.
- Intermediate producers:
  - linear labor technology  $y = zl$ ,
  - idiosyncratic AR(1) productivity  $\log z' = \rho \log z + \sigma_\varepsilon \varepsilon$ ,
  - menu cost  $f$  on price changes:  $V(s) = \max\{V_N(s), V_A(s)\}$ ,
  - state-dependent exit hazard  $\lambda(s)$ ; pricing internalizes survival through  $1 - \lambda(s')$ .
- Entry replaces exit one-for-one; entrants are smaller and price flexibly on arrival.
  - Entrants have lower productivity:  $G_e(z) \propto G(z) \cdot \exp(-\tau \log z)$ ,  $\tau \geq 0$

$s = (p_{-1}, z, S)$ : inherited price, productivity, aggregate state  $S = (P, Y, W, \mu)$ . Log preferences give  $W_t = \chi S_t$ : marginal cost moves with nominal demand.

## Exit hazard

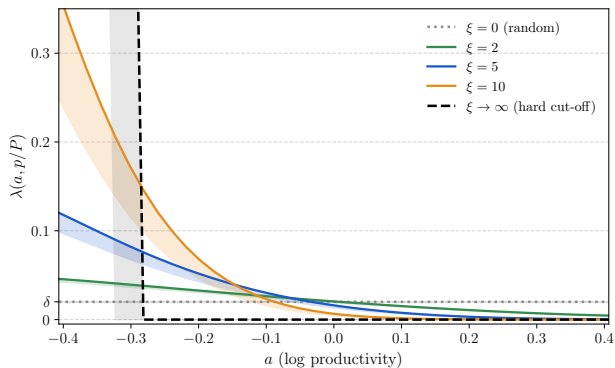
- The hazard is an exponential tilt of the value distribution, normalized to flow  $\delta$ :

$$\lambda(s) = \delta \cdot \frac{\exp(-\xi V(s))}{\mathbb{E}_\mu[\exp(-\xi V)]}, \quad \mathbb{E}_\mu[\lambda] = \delta$$

- Two parameters:  $\delta$  fixes the total exit rate,  $\xi \geq 0$  the elasticity of selection.
- Same  $\delta$ , different  $\xi$ : same total exit, the economies differ only in who leaves.

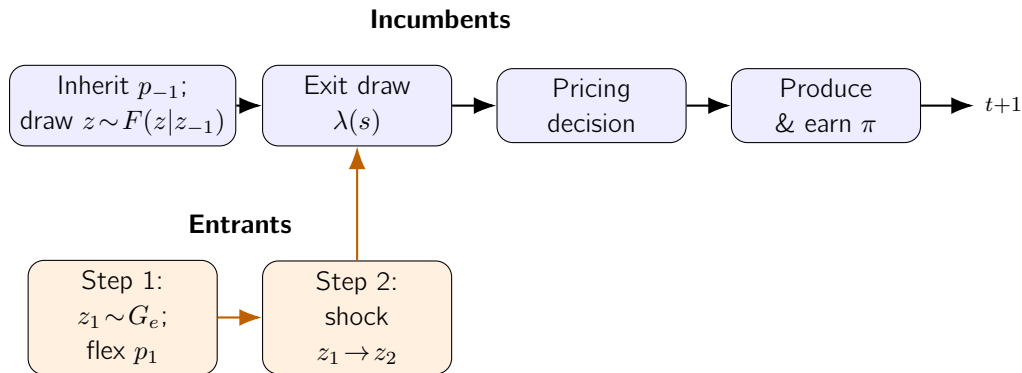
Two limits:  $\xi = 0$  is iid random exit at rate  $\delta$ ;  $\xi \rightarrow \infty$  concentrates the hazard on the lowest- $V$  firms, the cutoff rule of [Hopenhayn \(1992\)](#).

# Exit hazard by firm productivity



- Band: hazard range inside the inaction region; not adjusting can lower exit risk.

# Timing



$$\mu'(s') = \int (1 - \lambda(s)) Q(s, s') d\mu(s) + M \nu_e(s'), \quad M = \int \lambda d\mu$$

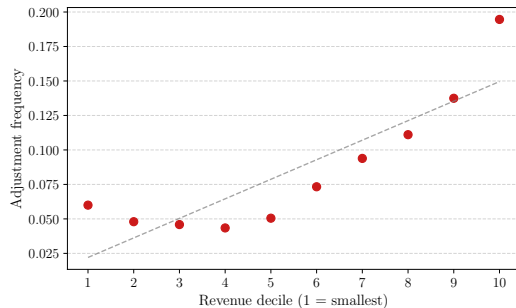
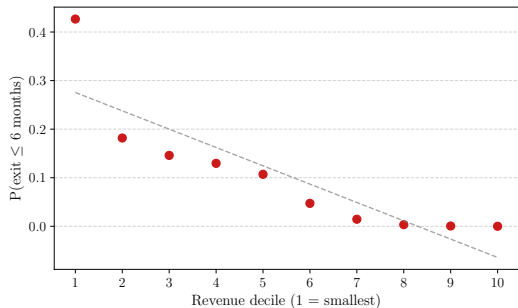
$Q$ : transition from pricing policy, productivity draw, aggregate state.  $\nu_e$ : entrant distribution.

# Calibration

| Parameter                    | Description                  | Value  | Target                                        | Data | Model |
|------------------------------|------------------------------|--------|-----------------------------------------------|------|-------|
| $f$                          | Menu cost                    | 0.155  | Frequency of price changes                    | 0.09 | 0.09  |
| $\delta$                     | Average exit rate            | 0.020  | Monthly exit rate                             | 0.02 | 0.02  |
| $\xi$                        | Exit selectivity             | 20     | Rel. size of exiting firms                    | 0.66 | 0.64  |
| $\tau$                       | Entry selectivity            | 5      | Rel. size of entrants                         | 0.67 | 0.69  |
| $\sigma_\varepsilon$         | St. dev. productivity        | 0.086  | Mean $\Delta \log p$                          | 0.04 | 0.04  |
| <i>Untargeted moments</i>    |                              |        |                                               |      |       |
| —                            | —                            | —      | Mean $ \Delta \log p $                        | 0.10 | 0.21  |
| —                            | —                            | —      | St. dev. $\Delta \log p$                      | 0.11 | 0.21  |
| —                            | —                            | —      | Fraction upward changes                       | 0.63 | 0.62  |
| <i>Externally calibrated</i> |                              |        |                                               |      |       |
| $\bar{\pi}$                  | Trend nominal demand         | 0.0028 | <a href="#">Nakamura and Steinsson (2010)</a> |      |       |
| $\sigma_S$                   | St. dev. nominal demand      | 0.0065 | <a href="#">Nakamura and Steinsson (2010)</a> |      |       |
| $\rho$                       | Productivity autocorrelation | 0.83   | Estimated                                     |      |       |
| $\beta$                      | Discount factor              | 0.96   | Externally calibrated                         |      |       |
| $\theta$                     | Demand elasticity            | 4      | Externally calibrated                         |      |       |
| $\chi$                       | Labor disutility             | 1      | Externally calibrated                         |      |       |

# The empirical facts, untargeted

- The calibration targets only aggregates and relative sizes, distributions are untargeted:



## Selection amplifies the response to a nominal shock

- Benchmark ( $\xi = 20$ ,  $\tau = 5$ ) vs. No selection ( $\xi = 0$ ,  $\tau = 0$ ), both at  $\delta = 0.02$ .
- CIR: one-year cumulated output response to a one-time shock to nominal spending.

| Model        | CIR (1 year) | vs. No selection |
|--------------|--------------|------------------|
| Benchmark    | 0.178        | +14.6%           |
| No selection | 0.155        | —                |
| No churn     | 0.156        | +0.5%            |

- Same total exit, same menu cost and pricing moments: +14.6% from composition.
- Long stochastic simulation: the variance of output is 15% higher under selection.

Solved along a deterministic perfect-foresight transition that tracks  $\{V_t, \mu_t, \lambda_t, \pi_t\}$  explicitly; the response is differenced from the contemporaneous no-shock path.

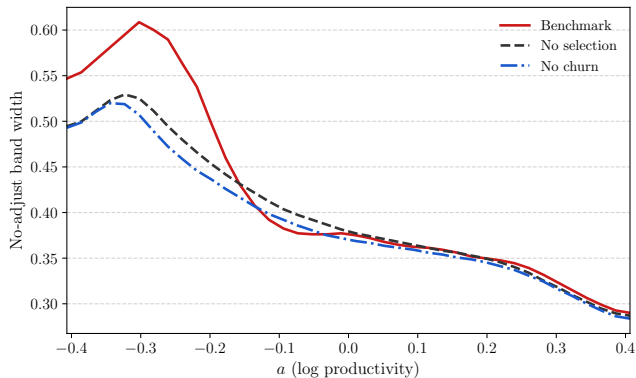
► CIR

► IRF backward-forward shooting algorithm

► Moments price changes distribution

►  $\text{Var}(Y)$

## Option value channel: exit risk widens inaction bands



- A high hazard shortens the horizon: paying the menu cost is worth less at low  $z$ .

## Censoring channel: two exits buy different amounts of flexibility

- **N.B.: CIR  $\approx$  amount of mispricing in the economy.**
- Every exit is replaced with a freshly priced entrant. What differs is the price it deletes.

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  - The entrant corrects that price early: random turnover buys aggregate flexibility.

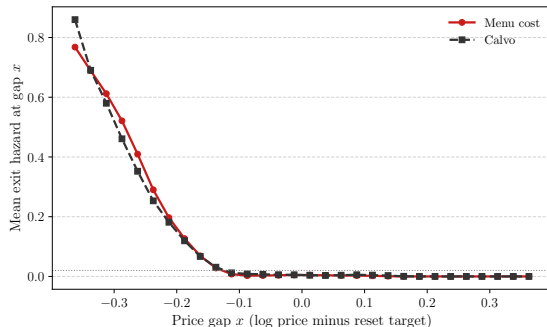
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  - The entrant corrects that price early: random turnover buys aggregate flexibility.
- Selective exit hits firms at the adjustment margin, whose correction was already coming.
  - The entrant posts the correction the incumbent was about to make: nothing changes.
- Reduced form sufficient condition: exit concentrated at the margins amplifies the CIR.

► Sufficient condition

At a fixed flow, selection withdraws exit from the interior: the subsidy switches off.

## Selective exit lands on firms about to adjust



- The measured hazard is negligible over the interior and steep where firms are underpriced.
- Random exit spreads  $\delta$  evenly, so it mostly deletes interior, persistent mispricing.

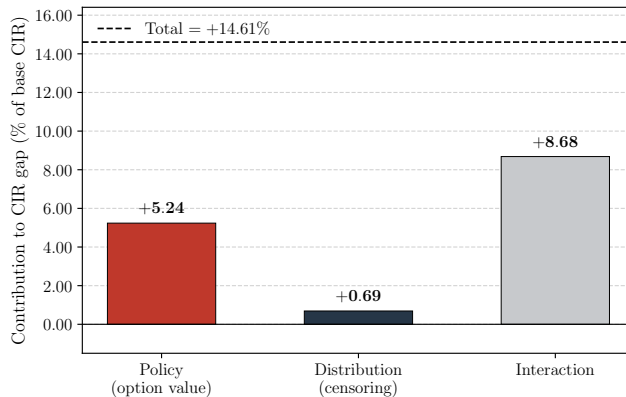
► Missing mass

► Who adjusts

► Phantom cohort

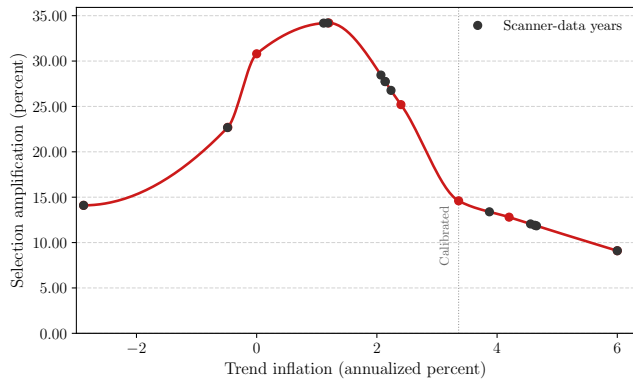
► Kernel

## CIR decomposition: complementarity between the channels

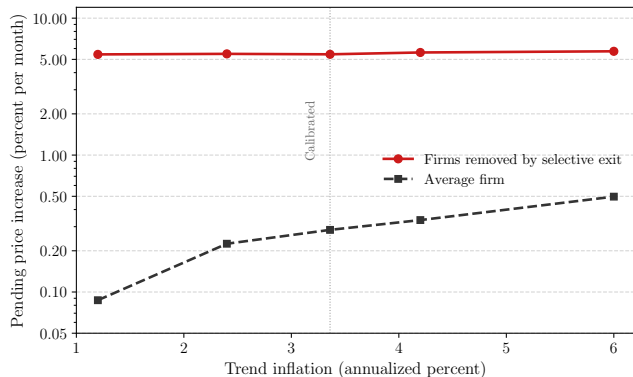


- Counterfactuals: firms price under a perceived hazard, mass exits under a realized one.

## Amplification is largest at low inflation

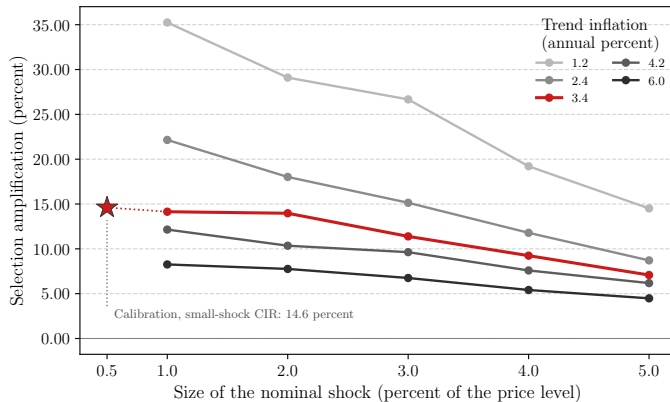


## Why the amplification dies at high inflation



- The firm exit removes is always about to raise its price; at low trend, the average firm is not.
- As trend inflation rises the average firm becomes more likely to increase prices.

# Large shocks work like high trend inflation



- A large shock releases at once the stock of mispricing selection has built.

## Calvo: selection still amplifies, by two percent

- Replace the menu cost with a Calvo lottery at  $\alpha = 0.088$ ; hazard and entry unchanged.
- One-year CIR: 0.552 to 0.565, a 2.4% amplification against 15% under the menu cost.
- Censoring flips to dampening: the lottery is memoryless, so exit deletes the largest gaps.
- The option value moves to the reset price: doomed firms target the static optimum.

► Model

► Decomposition

► Reset prices

► Moments

# Conclusion

- Selection on exit is a margin of monetary transmission.
  - Same exit rate, same pricing moments, 15% larger real effects; 35% at low inflation.
- Turnover acts as repricing technology; selection switches the flexibility subsidy off.
- Calibrations that treat exit as random miss part of the transmission.
- The margin matters most in ordinary low-inflation environments.

# Thank you!



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# Appendix slides

## The reduced model

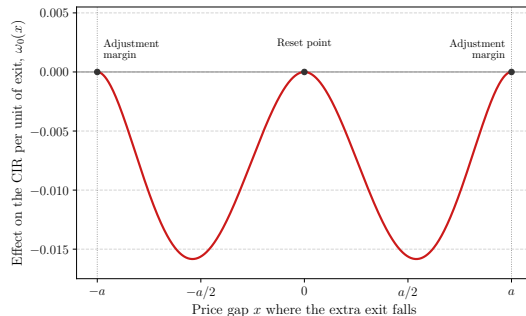
- Collapse the firm to its price gap  $x$ ; between adjustments  $dx = -\bar{\pi} dt + \sigma dW$ .
- Reset to 0 at the band edges  $\pm a$ ; exit at hazard  $\lambda(x)$ , entrant born at gap zero.
- At zero trend inflation, for any hazard and any band (fixed flow  $\mathbb{E}_f[\lambda] = \delta$ ):

$$\text{CIR} = \frac{\text{Var}_f(x)}{\sigma^2}, \quad \text{CIR}(\lambda_u) = \frac{a^2}{6\sigma^2} - \frac{a^4}{60\sigma^4} \lambda_u + O(\lambda_u^2)$$

- The shock survives in the standing spread of wrong prices: random churn subsidizes flexibility.

$x$ : posted price minus the free-adjustment price.  $f$ : stationary gap distribution.  $\lambda_u$ : uniform hazard.  $\sigma$ : volatility of gap innovations.

## The price of exit at each point of the cross-section



- One unit of exit at gap  $x$  moves the CIR by  $\omega_0(x) = -x^2(a - |x|)^2(a + |x|)/(6\sigma^4 a^2) \leq 0$ .
- Zero at the reset point and at the margins, strictly negative between: exit at the margin is free.
- At a fixed flow, hazard at the margins spares the interior: selection amplifies.

## When selection amplifies

### Condition M (margin-concentrated exit)

Relative to the uniform hazard with the same exit flow,

$$\int_{-a}^a \omega_0(x) (\lambda_s(x) - \lambda_u(x)) dx > 0.$$

### Theorem (selection amplifies)

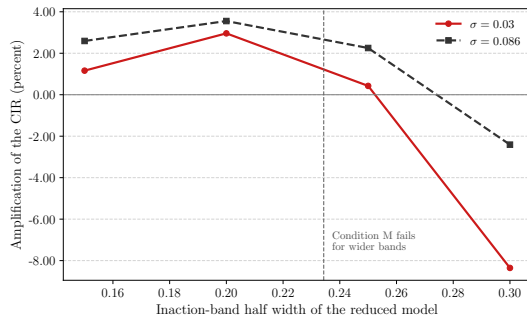
Under Condition M, at an equal exit flow and small intensity, the selective economy has the larger cumulative output response:  $CIR_s > CIR_u$ .

- Exit must concentrate near the band edges and spare the interior; no functional form.

► On the measured hazard

► Back

## The theorem on the measured hazard, and its boundary



- Feeding the measured hazard into the reduced model: 1 to 4% amplification from placement alone.
- Widening the band past 0.23 flips the sign: placement turns into dampening.

## Data: cleaning and price construction

- Non-promotion transactions only; unit-value prices in  $\text{€}[0.20, 25]$ , cent-rounded.
- Posted price: monthly mode per cell, kept with  $\geq 20$  transactions, modal share  $\geq 50\%$ .
- Lifecycles: span  $\geq 12$  months, density  $\geq 1/2$ , longest gap  $\leq 6$  months.
  - Left-censored excluded from entry, right-censored from exit.
- V-filter removes unflagged sales; change if  $0.001 \leq |\Delta \log p| \leq \log(10/3)$ .

**Table: Descriptive statistics.**

| Complete panel                 |                | Value       |
|--------------------------------|----------------|-------------|
| Sample period                  | 2008m1–2020m12 |             |
| Transactions (non-promotion)   |                | 163,524,586 |
| Recorded purchase value (mn €) |                | 341         |
| Unique products (UPCs)         |                | 999,062     |
| Brands                         |                | 20,404      |
| Retail chains                  |                | 231         |
| Product categories (groups)    |                | 719 (86)    |

Notes: The complete panel counts all non-promotion transactions, 2008–2020.

## Whose price changes: product or chain?

- A chain posts one price nationwide: same price in every province 98% of the time.
- If a chain changes a product's price, another chain follows within a month 63%.
  - Placebo with shifted timing: 14%; matched changes agree in sign 94%.
- I read a price change as coming from the product, not the retailer.

► Table

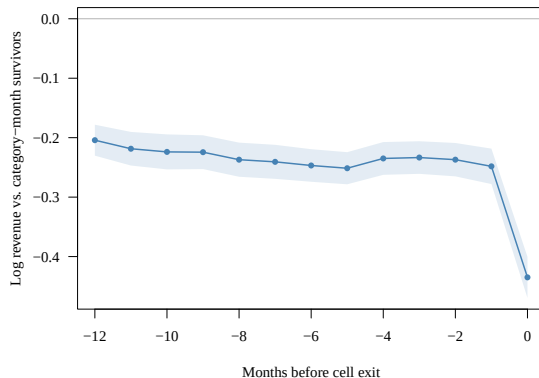
► Back

Table: Geography of price setting.

|                                            | Rate | Observations |
|--------------------------------------------|------|--------------|
| Same price in every province, within chain | 0.98 | 51,907       |
| Synchronization, $\pm 1$ month             | 0.63 | 4,778        |
| Same province only                         | 0.62 | 799          |
| Chain level                                | 0.55 | 2,067        |
| Same month only                            | 0.46 | 4,778        |
| Placebo: random time shifts                | 0.14 | 4,778        |
| Same sign among matched changes            | 0.94 | 3,446        |

Notes: The first row covers the 51,907 product-chain-months of the pricing sample observed in two or more provinces, and reports the share in which the modal posted price is identical in every one of them; the mean standard deviation of the log price across provinces is 0.001, against 0.028 across chains in the same product-month (4,242 product-months). The remaining rows use the 185 products of the pricing sample sold in two or more chains (22,165 cell-months with a defined change indicator). An opportunity is a pair (product, month) in which one chain changes the product's price and a different chain sells the product with a defined change indicator in that month; a match occurs if the second chain changes the price within one month. Same province restricts the pair to cells in the same province. Chain level aggregates cells to the product-chain-month before pairing. The placebo shifts each cell's change months by a random offset of between three and 24 months (20 draws, mean reported). The last row is the share of matched cross-chain change pairs in which the two changes have the same sign.

## Revenue before exit

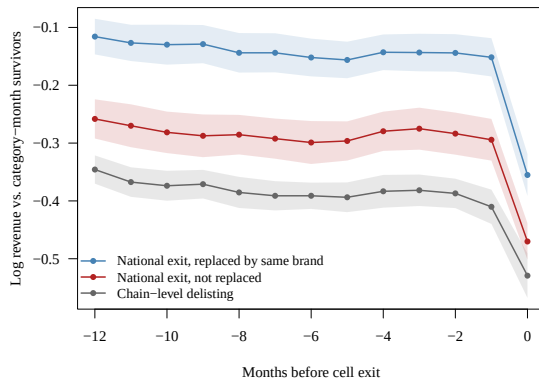


- Exiting products sell 20% below category-month peers a year before they disappear.

► By type

► Back

## Revenue before exit, by type of exit



- Replaced exits decline as clearly as unreplaced ones: brands retire dying variants.

# What barcode exit measures

Table: What barcode exit measures.

|                                                         | Share              |
|---------------------------------------------------------|--------------------|
| National product exit within 6 months, revenue-weighted | 0.82               |
| products sold in 2+ chains                              | 0.70               |
| Same-brand entrant in same category, $[-3, +6]$ months  | 0.62               |
| at a random month of the same brand-category (placebo)  | 0.50               |
| excess over placebo                                     | 0.12               |
| closest entrant: same / smaller / larger pack           | 0.34 / 0.30 / 0.34 |

Notes: Sample: valid lifecycles of the turnover sample (Table 1), interior months; 96,915 cell exits, of which 27,263 national product exits with an identified manufacturer brand. A national product exit occurs when the UPC leaves every chain and province in the panel within six months of the cell exit; shares are weighted by lifecycle revenue. The replacement rows report the share of national product exits followed by a new barcode of the same brand in the same category within three months before to six months after the exit; the placebo evaluates the same window at a random interior month for the same brand-category, averaged over ten draws. Pack size compares volume per unit of the exiting product and its closest replacement, when both share a measurement unit.

- Excess same-brand replacement tied to the exit itself: twelve percentage points.

## Pricing does not change on the way out

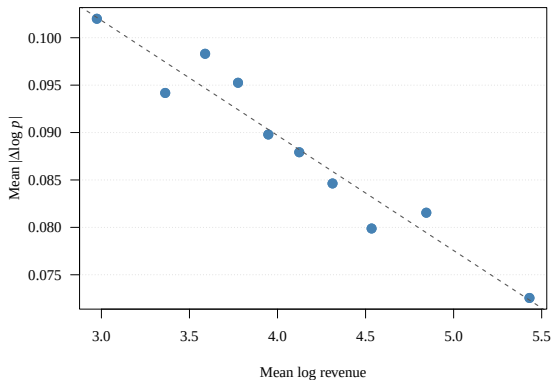
**Table: Adjustment frequency by months before exit.**

| Months before exit          | Frequency | <i>N</i> |
|-----------------------------|-----------|----------|
| 1–3                         | 0.079     | 5,825    |
| 4–6                         | 0.100     | 4,896    |
| 7–12                        | 0.088     | 10,823   |
| 13 or more                  | 0.079     | 76,701   |
| Censored (no observed exit) | 0.079     | 70,834   |

Notes: Mean monthly adjustment probability on the incumbent panel of the pricing sample, by months between the observation and the product's exit from the sample. The censored row collects products whose exit is never observed.

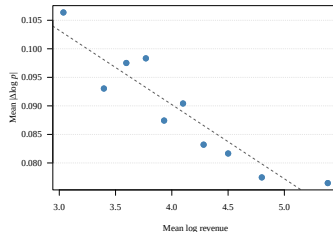
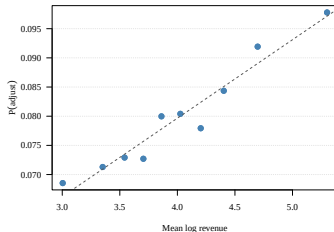
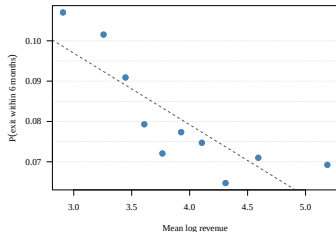
- What distinguishes exiting products is their level of revenue, not their timing.

## Size of price changes by revenue decile



- Conditional on adjusting, smaller products move by about ten log points, larger by seven.

## Robustness: trailing twelve-month size proxy



- Also robust to: national and brand exits, sample halves, quantity margin, within-brand.

► Back

## Persistence of idiosyncratic productivity

- Monthly prices carry measurement error; error sits in the variance, not the autocovariances:

$$C_0 = \text{Var}(x^*) + \text{Var}(u), \quad C_k = \rho^k \text{Var}(x^*) \quad (k \geq 1)$$

- Ratios of successive autocovariances recover  $\rho$  whatever the error:

$$\rho = \frac{C_2}{C_1} = \frac{C_3}{C_2}$$

- The two ratios give 0.830 and 0.833; I set  $\rho = 0.83$ .
- A regression on the first lag gives 0.512: 38% of price movement is error.

## Incumbent's problem

- Value, conditional on surviving this period's exit draw:

$$V(s) = \max\{V_N(s), V_A(s)\}$$

- Keep the inherited price, or pay the menu cost and reset:

$$\begin{aligned} V_N(p_{-1}, z, \mathcal{S}) &= \pi(p_{-1}, z, \mathcal{S}) + \mathbb{E}[\Xi(1 - \lambda(s')) V(p_{-1}, z', \mathcal{S}')] \\ V_A(z, \mathcal{S}) &= \max_p \left( \pi(p, z, \mathcal{S}) + \mathbb{E}[\Xi(1 - \lambda(s')) V(p, z', \mathcal{S}')] \right) \end{aligned}$$

- Flow profit carries the labor-denominated menu cost  $f$  when the price changes.

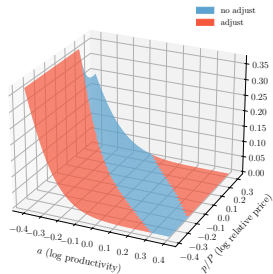
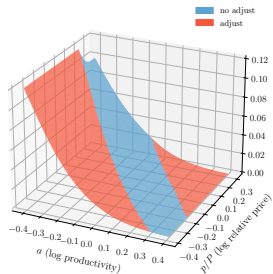
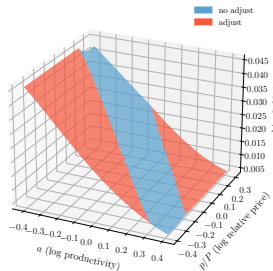
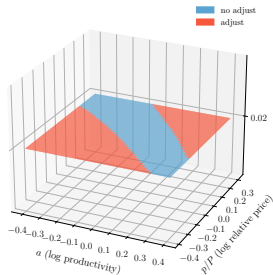
$\Xi$ : stochastic discount factor.  $\lambda(s')$ : next-period exit hazard, so pricing lowers exit risk through  $V$ .

- Entry is one-for-one with exit:  $M = \int \lambda(s) d\mu(s)$ .
- Two-step entry block within the period:
  1. Draw  $z_1 \sim G_e$ ; set the flexible price  $p_1 = \arg \max_p V_A(z_1, \mathcal{S})$ , no menu cost.
  2. Within-period shock  $\log z_2 = \rho \log z_1 + \sigma_\varepsilon \varepsilon$ : entrants start slightly mispriced.
- Entry distribution tilted toward low productivity:

$$G_e(z) \propto G(z) \cdot \exp(-\tau \log z), \quad \tau \geq 0$$

- $\tau = 0$ : entrants mirror incumbents;  $\tau > 0$ : entrants are smaller, as in the data.

# Exit hazard in two dimensions



## Definition 1

A stationary recursive competitive equilibrium consists of values  $V_N, V_A$ , a price policy  $p(s)$ , an exit hazard  $\lambda(s)$ , an entrant distribution  $\nu_e$  and mass  $M$ , an invariant measure  $\mu$ , aggregates  $\{C, W/P\}$ , and a perceived law of motion  $G$ , such that

- 1  $C = S/P$  and  $W/P = \chi C$  solve the household problem;
- 2 given  $G, W/P$ , and  $\lambda$ , the policy  $p(s)$  solves the incumbent Bellman;
- 3 the hazard satisfies its normalization against  $\mu$ ;
- 4  $\nu_e$  follows the entry block and  $M = \int \lambda d\mu$  (one-for-one replacement);
- 5  $\mu$  is invariant under the law of motion;
- 6 goods and labor markets clear;
- 7  $G$  is consistent with the inflation generated by the firm sector.

## Krusell-Smith solution algorithm

Linear rule  $G(\mathcal{S}) = \bar{\pi} + b_1(\mathcal{S} - s^*)$ , with  $s^*$  the aggregate-state fixed point.

- 1 Guess  $(s^*, b_1)$ .
- 2 VFI given  $G \rightarrow$  price policy  $p(s)$ .
- 3 Solve for the stationary  $\mu$  given  $p$  and  $G$ .
- 4 For each  $\mathcal{S}$ , compute the model-implied inflation

$$\hat{G}(\mathcal{S}) = \frac{1}{1-\theta} \log \mathbb{E}_\mu \left[ e^{(1-\theta)p(s)} \mid \mathcal{S} \right].$$

- 5 Fit  $\hat{G}$  linearly by WLS with weights  $\propto$  mass  $\rightarrow$  updated  $(s^*, b_1)$ .
- 6 Iterate to  $\|G_{\text{new}} - G_{\text{old}}\|_\infty < \varepsilon$ .

## Unconditional variance of output

- Nominal demand follows the random walk with drift; draw  $\eta_t \sim N(0, \sigma_\eta^2)$ .
- Given the converged rule  $G$ , iterate forward

$$P_{t+1} = P_t + G(S_t - P_t), \quad Y_t = S_t - P_{t+1}$$

- Simulate 25,000 periods after burn-in and report the sample variance of  $Y_t$ .
- Selection raises  $\text{Var}(Y)$  by about 15%; random churn alone moves it by 4%.

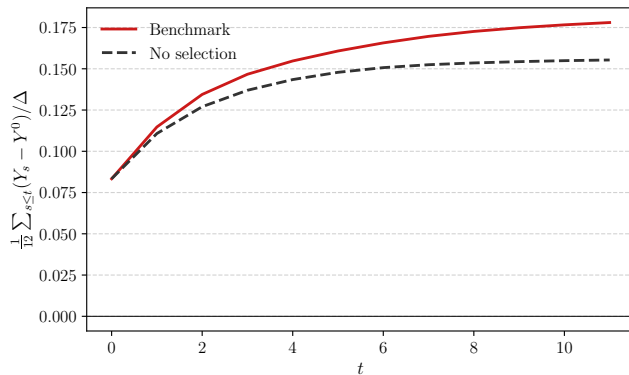
## Backward-forward shooting

Outer iteration on the inflation path  $\{\pi_t\}_{t=0}^{T-1}$ :

- 1 Guess  $\{\pi_t\}$ ; recover  $\{Y_t\}$  from  $Y_{t+1} = Y_t + \bar{\pi} - \pi_t$ .
- 2 Backward pass:  $V_T = V^{ss}$ ; solve the Bellman for  $t = T - 1, \dots, 0$  given  $\lambda_t$ .
- 3 Forward pass:  $\mu_0 = \mu^{ss}$ ; iterate  $\mu_{t+1} = \mathcal{T}_t[\mu_t]$ .
- 4 Update  $\lambda_t$  from the contemporaneous  $(V_t, \mu_t)$ .
- 5 Compute  $\pi_t^{\text{new}}$  from the CES condition; damped update until convergence.

Boundary conditions from the deterministic stationary equilibrium; the reported IRF is differenced from the contemporaneous no-shock path,  $\psi_t = (\log Y_t - \log Y_t^0)/\Delta$ .

## Cumulative response of output



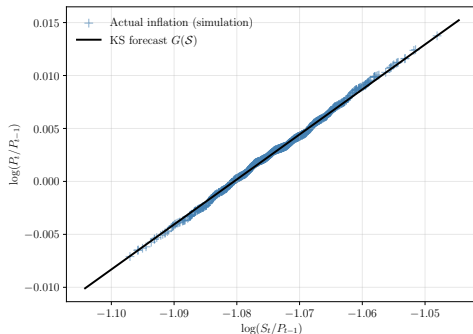
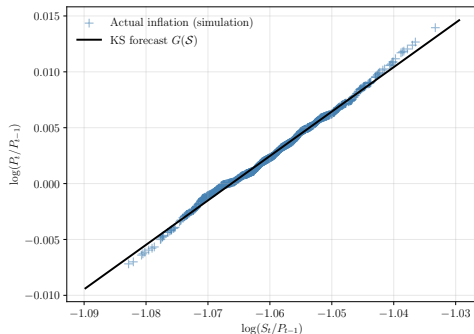
## Stationary moments and output variance

Table: Pricing moments and unconditional output variance.

| Model        | $\delta$ | $\xi$ | $\tau$ | Frequency | Kurtosis | $\text{Var}(Y) \times 10^5$ |
|--------------|----------|-------|--------|-----------|----------|-----------------------------|
| Benchmark    | 0.02     | 20    | 5      | 0.086     | 1.41     | 2.36                        |
| No selection | 0.02     | 0     | 0      | 0.088     | 1.32     | 2.05                        |
| No churn     | 0.00     | 0     | 0      | 0.092     | 1.31     | 1.97                        |

- Kurt/freq: sufficient statistic for small nominal shocks ([Alvarez et al., 2016](#)).

## Forecast errors: linear rule

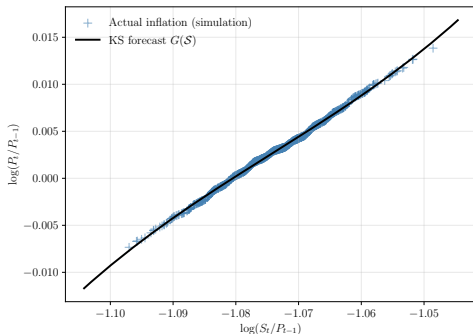
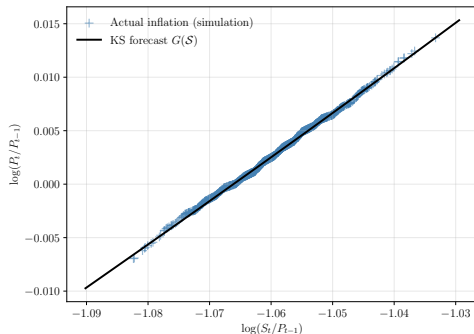


- Benchmark (left) bows at the support boundaries; No selection (right) is linear.

► Cubic rule

► Back

## Forecast errors: cubic rule



- The bow disappears; the forecast-error dispersion drops by a third in the Benchmark.

► Moments

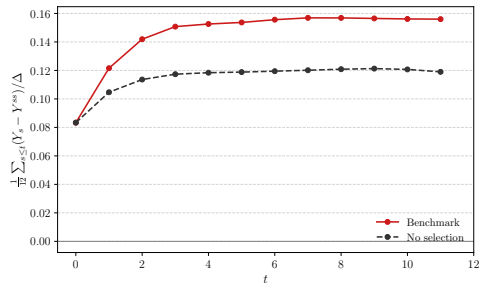
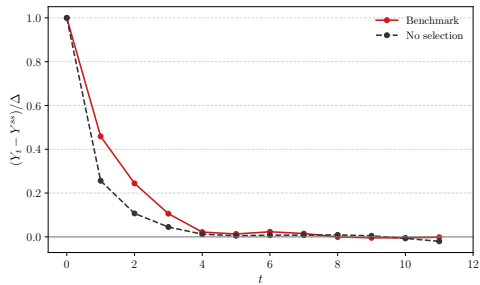
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Table: Stationary moments under linear vs. cubic forecasting rule.

| Model        | Linear rule |          |                             | Cubic rule |          |                             |
|--------------|-------------|----------|-----------------------------|------------|----------|-----------------------------|
|              | Frequency   | Kurtosis | $\text{Var}(Y) \times 10^5$ | Frequency  | Kurtosis | $\text{Var}(Y) \times 10^5$ |
| Benchmark    | 0.086       | 1.41     | 2.362                       | 0.086      | 1.39     | 2.190                       |
| No selection | 0.088       | 1.32     | 2.051                       | 0.088      | 1.32     | 1.988                       |
| No churn     | 0.092       | 1.31     | 1.967                       | 0.092      | 1.31     | 1.988                       |

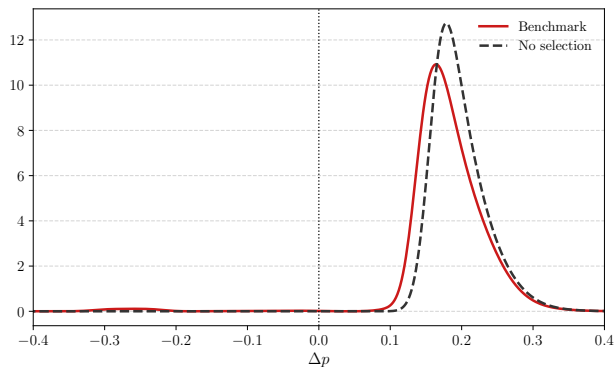
- Pricing moments unchanged; the variance gap shrinks from +15.4 to +10.3%.
- One-year CIRs differ by less than one percent across rules.

## IRF and CIR under the cubic rule



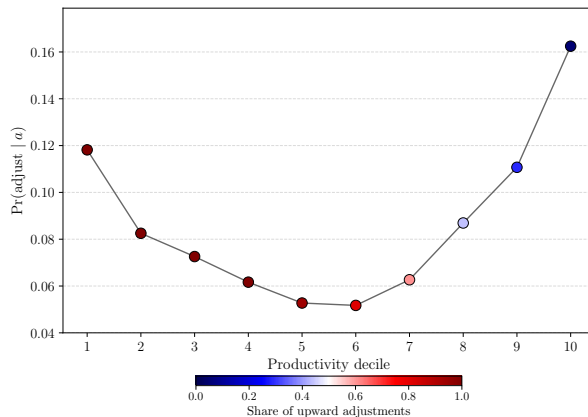
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## Missing mass: lowest productivity quartile



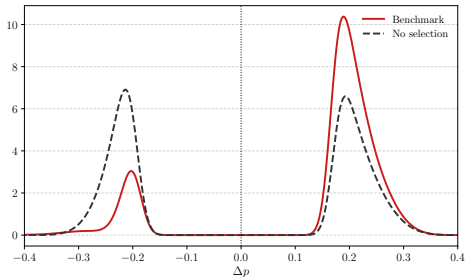
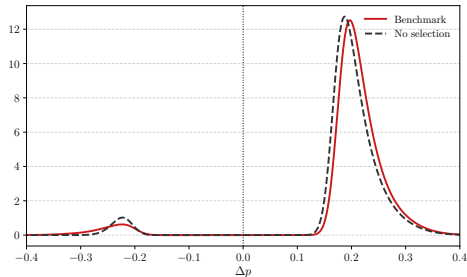
- The mass selection removes from Q1 is the upward resets exit intercepts.

## Who adjusts, by productivity



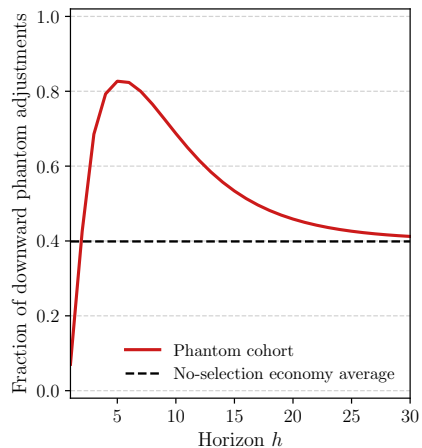
- Adjustment concentrates in the tails; selective exit removes the low- $z$ , high-adjustment tail.

## Missing mass: middle quartiles



- Less mass at moderate downward adjustments: the mean-reversion leg goes missing too.

## Phantom cohort vs. uniform cohort



- The selectively removed cohort adjusts far more than a uniformly drawn one.

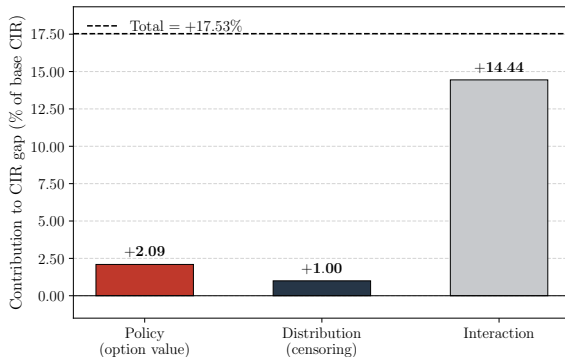
## Decomposition counterfactuals

- The hazard does two jobs: it discounts continuation values, and it deletes mass.
- I separate them, holding the flow  $\delta$  fixed:
  - firms price under a perceived hazard  $\lambda_B$  in the Bellman equation,
  - the cross-section evolves under a realized hazard  $\lambda_F$  in the law of motion:

$$\lambda_j(s) = \delta \cdot \frac{\exp(-\xi_j V(s))}{\mathbb{E}_\mu[\exp(-\xi_j V)]}, \quad j \in \{B, F\}$$

- Each configuration  $(\xi_B, \xi_F, \tau)$  is solved and shocked; one channel at a time.
  - Option value:  $(20, 0)$ ; censoring:  $(0, 20)$ ; interaction: residual to the full  $(20, 20)$ .

## CIR decomposition: entry-only to exit-only



- Moving selectivity from entry to exit raises the CIR by about 18%.
- The standalone bars nearly vanish: selection works only as a package.

## Moment decomposition: menu-cost model

Table: Decomposition of selection effects, menu-cost model.

| Moment                                                                                                       | $M_0$  | $M_1$  | $M_1 - M_0$ | $\Delta_{\text{policy}}$ | $\Delta_{\text{dist}}$ | $\Delta_{\text{int}}$ |
|--------------------------------------------------------------------------------------------------------------|--------|--------|-------------|--------------------------|------------------------|-----------------------|
| <i>Pair 1 — No-selection (<math>\xi = 0, \tau = 0</math>) vs Benchmark (<math>\xi = 20, \tau = 5</math>)</i> |        |        |             |                          |                        |                       |
| mean $\Delta \log p$                                                                                         | +0.032 | +0.042 | +0.010      | -0.008                   | +0.015                 | +0.003                |
| kurtosis $\Delta \log p$                                                                                     | 1.318  | 1.415  | +0.097      | -0.022                   | +0.124                 | -0.006                |
| frequency (pp)                                                                                               | 8.785  | 8.581  | -0.204      | +0.050                   | -0.193                 | -0.061                |
| <i>Pair 2 — entry-only (<math>\xi = 0, \tau = 5</math>) vs exit-only (<math>\xi = 20, \tau = 0</math>)</i>   |        |        |             |                          |                        |                       |
| mean $\Delta \log p$                                                                                         | +0.012 | +0.052 | +0.040      | -0.030                   | +0.047                 | +0.023                |
| kurtosis $\Delta \log p$                                                                                     | 1.203  | 1.508  | +0.305      | +0.016                   | +0.366                 | -0.076                |
| frequency (pp)                                                                                               | 8.746  | 8.496  | -0.250      | -0.963                   | -0.140                 | +0.853                |

## Moment decomposition: Calvo model

Table: Decomposition of selection effects, Calvo model.

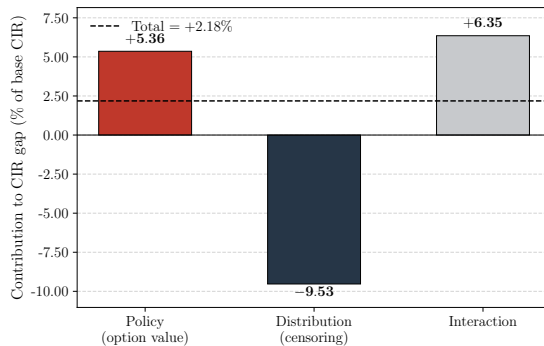
| Moment                                                                                                       | $M_0$  | $M_1$  | $M_1 - M_0$ | $\Delta_{\text{policy}}$ | $\Delta_{\text{dist}}$ | $\Delta_{\text{int}}$ |
|--------------------------------------------------------------------------------------------------------------|--------|--------|-------------|--------------------------|------------------------|-----------------------|
| <i>Pair 1 — No-selection (<math>\xi = 0, \tau = 0</math>) vs Benchmark (<math>\xi = 20, \tau = 5</math>)</i> |        |        |             |                          |                        |                       |
| mean $\Delta \log p$                                                                                         | +0.026 | +0.026 | -0.001      | +0.059                   | -0.054                 | -0.006                |
| skewness $\Delta \log p$                                                                                     | +0.377 | +0.205 | -0.173      | +0.450                   | -0.580                 | -0.044                |
| kurtosis $\Delta \log p$                                                                                     | 3.667  | 4.795  | +1.128      | +0.612                   | +0.384                 | +0.132                |
| <i>Pair 2 — entry-only (<math>\xi = 0, \tau = 5</math>) vs exit-only (<math>\xi = 20, \tau = 0</math>)</i>   |        |        |             |                          |                        |                       |
| mean $\Delta \log p$                                                                                         | +0.018 | +0.035 | +0.016      | +0.026                   | -0.003                 | -0.007                |
| skewness $\Delta \log p$                                                                                     | +0.342 | +0.701 | +0.359      | +0.354                   | +0.161                 | -0.156                |
| kurtosis $\Delta \log p$                                                                                     | 3.692  | 4.626  | +0.934      | +0.825                   | +0.092                 | +0.018                |

- No menu cost in flow profits; adjustment arrives with probability  $\alpha$ .
- Values:

$$V_N(p, z, \mathcal{S}) = \pi(p, z, \mathcal{S}) + \mathbb{E}[\Xi(1 - \lambda(s'))((1 - \alpha)V_N(p, z', \mathcal{S}') + \alpha V_A(z', \mathcal{S}'))]$$
$$V_A(z, \mathcal{S}) = \max_{p'} V_N(p', z, \mathcal{S})$$

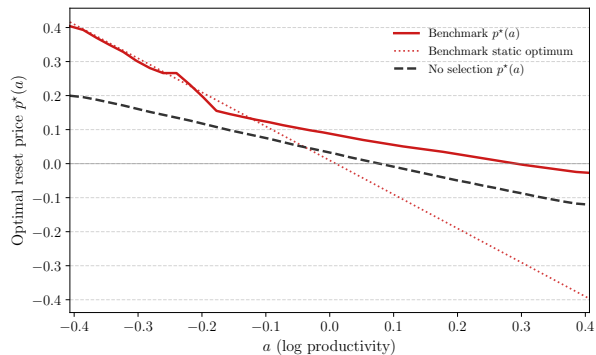
- Household, productivity, exit hazard, and entry block unchanged.
- In the timing, the pricing decision becomes an exogenous Bernoulli draw.

## CIR decomposition in the Calvo model



- One-year CIRs: no selection 0.552, policy only 0.582, distribution only 0.500, benchmark 0.565.

## The option value moves to the reset price



- Firms facing near-certain exit stop pricing dynamically and target the static optimum.

Table: Calvo model: pricing moments under selective exit, baseline calibration.

|              | $\xi$ | $\tau$ | Frequency | $\sigma(\Delta \log p)$ | skewness | kurtosis |
|--------------|-------|--------|-----------|-------------------------|----------|----------|
| No selection | 0     | 0      | 0.088     | 0.081                   | +0.377   | 3.667    |
| Benchmark    | 20    | 5      | 0.088     | 0.084                   | +0.205   | 4.795    |

- Frequency identical by construction; the action is in the shape of  $\Delta \log p$ .

## Sensitivity to persistence: Calvo

Table: Calvo model: pricing moments by persistence and selection.

| Parameterisation                          | $\xi$ | $\tau$ | Frequency | $\sigma(\Delta \log p)$ | $\text{skew}(\Delta \log p)$ | Kurtosis |
|-------------------------------------------|-------|--------|-----------|-------------------------|------------------------------|----------|
| $\rho = 0.85, \sigma_\varepsilon = 0.086$ | 0     | 0      | 0.09      | 0.08                    | +0.38                        | 3.67     |
|                                           | 20    | 5      | 0.09      | 0.08                    | +0.20                        | 4.79     |
| $\rho = 0.95, \sigma_\varepsilon = 0.051$ | 0     | 0      | 0.09      | 0.09                    | +0.52                        | 4.46     |
|                                           | 20    | 5      | 0.09      | 0.08                    | +0.43                        | 4.28     |

- At  $\rho = 0.95$  the Calvo kurtosis ranking reverses: the policy channel needs fast reversion.

Table: Menu-cost model: pricing moments by persistence and selection.

| Parameterisation                          | $\xi$ | $\tau$ | Frequency | $\sigma(\Delta \log p)$ | $\text{skew}(\Delta \log p)$ | Kurtosis |
|-------------------------------------------|-------|--------|-----------|-------------------------|------------------------------|----------|
| $\rho = 0.85, \sigma_\varepsilon = 0.086$ | 0     | 0      | 0.09      | 0.21                    | -0.41                        | 1.32     |
|                                           | 20    | 5      | 0.09      | 0.21                    | -0.45                        | 1.41     |
| $\rho = 0.95, \sigma_\varepsilon = 0.051$ | 0     | 0      | 0.06      | 0.17                    | -0.53                        | 1.39     |
|                                           | 20    | 5      | 0.06      | 0.16                    | -0.64                        | 1.56     |

- The menu-cost kurtosis gap survives and slightly widens: +0.10 to +0.17.